Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/05/2011
PI Name:	Rose, Raphael Ph.D.		
Project Title:	A Multi-Media, Computer-Based, Self-Directed, Autonomous, Stress and Anxiety-Management Countermeasure		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-1563	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay (Dartmouth College) Craske, Michelle (University of California, Los Angeles) Beven, Gary (NASA JSC) Glover, Dorie (Dorothy) (University of California, Los Angeles)		
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Task Description:	1. Original Project aims/objectives: Stress and anxiety-related problems before, during and after space missions can seriously compromise crew efficiency and safety. The Institute of Medicine report of Earth-analog environments (e.g., Antarctica) suggests a prevalence rate of serious behavioral problems of up to 13% per person per year, highlighting the potential behavioral problems that could occur on a 3-year mission. Our project has two specific aims. The first aim is to develop a self-directed, autonomous, interactive multimedia program to train astronauts how to recognize, assess, detect, prevent, and manage stress on extended spaceflights. The countermeasure would be principally used in pre-flight training, with booster sessions or interventions available on orbit as needed. In this model of implementation, the skills learned prior to a mission would help to "inoculate" fliers to severe stress reactions, thereby enhancing flight-task performance and crew health, safety and efficiency, and build resilience. The countermeasure, called "Stress Management and Resilience Training for Optimal Performance" or SMART-OP, will incorporate empirically supported stress anxiety management strategies delivered by interactive media. The second aim is to evaluate the efficacy of the program with an astronaut-relevant sample in a randomized controlled trial. The final product will be a computer-based stress management and resilience training program that could be used both by flight crews and ground-personnel to prepare for stress during training, while on mission, or post-mission. 2. Key Findings: Our autonomous, computer-based, multimedia stress management and resilience training program is currently being evaluated in a randomized controlled efficacy trial at The University of California, Los Angeles (UCLA). Therefore, no outcome data has been analyzed. Data collection will be completed in July 2011. Thus far, our target enrollment sample (N=60) has been exceeded. Our sample, in this phase 1 efficacy trial, was chosen to be somewhat analogous to an astronaut population. We selected healthy individuals (no medical or psychiatric conditions) and work in a demanding and potentially stressful environment (i.e., law and business school). Interested study participants were screened to ensure they met study eligibility criteria. We administered the Mini International Neuropsychiatric Interview (MINI) to rule out any individuals who met full DSM-IV-TR (Diagnostic and Statistical Manual 4th Edition-Text Revision) criteria for a psychiatric diagnosis. Potential participants also had to be free of any significant medical conditions, and endorse a certain level of stress based on their score on a widely used self-report questionnaire (Perceived Stress Scale - PSS, Cohen, 1983). We selected a cutoff score of 0.5 SD above the mean score for a community norm on the PSS. We have had great interest in the study, with over 220 individuals contacting us. Our research team screened 164 of those individuals and 85 were deemed ineligible, while 79 were eligible. Of those who were ineligible, the most frequent reason was meeting full DSM-IV-TR diagnostic criteria for a psychiatric diagnosis or too low a stress level based on their PSS score. Nearly all prior research on stress management outcome studies were conducted with "clinical" samples (e.g., anxious or medically ill subjects) so our screening procedures are quite novel. Of those who were eligible, 61 pre-assessments have been completed and they were randomized to one of the two study groups; the SMART-OP group (N=33) who receive computerized stress management (CSM) or to an Attention Control (AC) (N=28) who watch videos and read information on stress management. To date, we have completed 30 post-assessments. A sub-sample (N=14) of SMART-OP user ratings of usefulness (1-7 scale with higher scores indicating greater usefulness) of program activities shows that they rated them very positively (M=5.37, SD= 1.2) and higher, on average, than the control group rated their activities (M=4.25, SD=1.56). While this is only a small sub-sample, it provides some initial positive feedback on the usefulness of our countermeasure. 3. Impact of key findings on hypotheses, technology requirements, objectives and specific aims of the original proposal: Since we have not yet completed our efficacy trial we do not have key findings to report. However, our process of reviewing the stress literature and meeting with experts at NASA and the US Military informed many of our decisions in terms of content and design of the program. SMART-OP is designed to train people who work in operational environments like space flight and military to deal with building resilience and mental fitness to the types of stressors that are often chronic in nature and stressors they are not specifically trained to address. That is, those "outside" of the direct scope of a mission (e.g., family problems, financial, health, and conflicts with colleagues). Our research led us to focus on pre and post mission stress, as data suggests that individuals are trained very well to handle/solve mission-related stressors (e.g., emergency procedures, combat tactics), but not specifically trained on how to handle problems at home or with a colleague that are less amenable to being "solved." We designed our program with the concept that if an individual is handling the stressors outside of the direct scope of his/her duties well, then they can be better able to focus on their job, handle the stress of their job well, and perform optimally. 4. Proposed research plan for the coming year: We plan to complete the efficacy trial, analyze data, and prepare findings for presentations and manuscripts. We also hope to have a small sample (N=5-10) of US Marine Pilots review the program and provide qualitative feedback on usefulness and validity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Stress Management and Resilience Training for Optimal Performance (SMART-OP), is a computer-based, self-guided multimedia, interactive, evidenced-based stress management and resilience training program. As far as we know, this program is the first of its kind. At the end of this study, we will not only have developed an evidenced-based stress management and resilience training countermeasure with potential value to the space program and other operational environments (e.g., Military, law enforcement, emergency responders), but also with some modifications and revisions, to the broader American public. Stress-related health and mental problems, and work-related stress and burnout are among the most common and costly health problems in the country. Stress also plays a significant etiological role in the onset of many physical and psychiatric disorders. In fact, the "Stress Diathesis Model" posits that stress in conjunction with a biological predisposition is responsible for the onset of many psychiatric problems like anxiety and depression. After efficacy studies are completed on SMART-OP, a potential next step would be larger-scale effectiveness studies to further identify in which sub-populations (e.g., submarines, Antarctic research stations, commercial ships, and oil rigs) and user settings (e.g., home, clinic, via Internet etc.) it is effective. Ultimately, the program may have benefit in helping to prevent mental and medical problems in the future by providing easy access to stress management skills that help build resilience and mental fitness and may help mitigate the onset of mental health and medical disorders. SMART-OP has the potential for widespread dissemination via Internet to provide stress-management training to individuals worldwide.

Task Progress:	Summary: SMART-OP (Stress Management and Resilience Training for Optimal Performance) is a self-guided, multimedia, interactive, computer-based, stress management and resilience training program based on evidenced-based cognitive-behavioral principles. The program is currently being evaluated in a randomized controlled efficacy trial with stressed but healthy graduate (Law and Business) students at the University of California, Los Angeles (N = 60) where it is being compared to an attention control (AC) group. Outcome is based on subjects' performance during a laboratory social stress test (Trier) where they give a speech, solve math problems, and complete a memory and non-verbal motor performance task in front of a small audience. Outcome measures include psychophysiological responses, biomarkers for stress (cortisol and salivary amylase), and self-report questionnaires. Program Production: We completed design and production of SMART-OP and a practice USB flashdrive for in between session practice. SMART-OP contains six 30 to 45-minute sessions that teach stress management skills in the areas of education, emotion/physiological regulation, cognitive flexibility, and behavioral actions to cope with stressful situations. The skills are taught in an interactive and engaging manner. For example, in the Biofeedback Challenge, users learn to improve heart-rate variability by playing a video race car game while getting feedback on their breathing rate. We created hundreds of pages of storyboard and scripts with detailed descriptions and instructions for our multimedia production company. We professionally shot/re-shot and edited well over a hundred hours of video and audio footage over the course of five months. We videotaped interviews with experts in the field of stress management (Drs. Michelle Craske and Don Meichenbaum), as well with individuals familiar with working in highly demanding fields like space exploration (Drs. Jay Buckey, Mike Barratt, and Gary Beven) and the military (Lt. Col. Eddie Fox). Lastly, the program was reviewed by nearly a dozen people to check for glitches and/or logic problems etc. Efficacy Trial: We began our randomized controlled efficacy trial. To date, 222 individuals expressed interest in participating in our study at UCLA. We screened 164 individuals and 79 of those met our study eligibility criteria (stressed but otherwise healthy). 61 individuals completed their pre-assessments and were randomized into the SMART-OP or AC groups, and 30 have completed their post-assessment. Presentations and Interest: Aspects of the program have been presented to Space Flight Resource Management at JSC; 3 International Conferences; 2 UCLA departmental research meetings; a customer agreement between HRP and the PI was created for sharing program content; and members from the US Marines, US Coast Guard, and Office of Rail Safety in Australia have expressed interest in future collaborations and applications of the program.
Bibliography Type:	Description: (Last Updated: 02/11/2021)
Awards	Rose RD. "Promotion to Associate Research Psychologist, Step II, July 2010." Jul-2010
Books/Book Chapters	Rose RD, Buckey JC, Craske MG, Cartreine JA. "Stress management and resilience training for optimal performance on long-duration space flight." Berkeley, Calif. : University of California Press, in press, 2011., May-2011